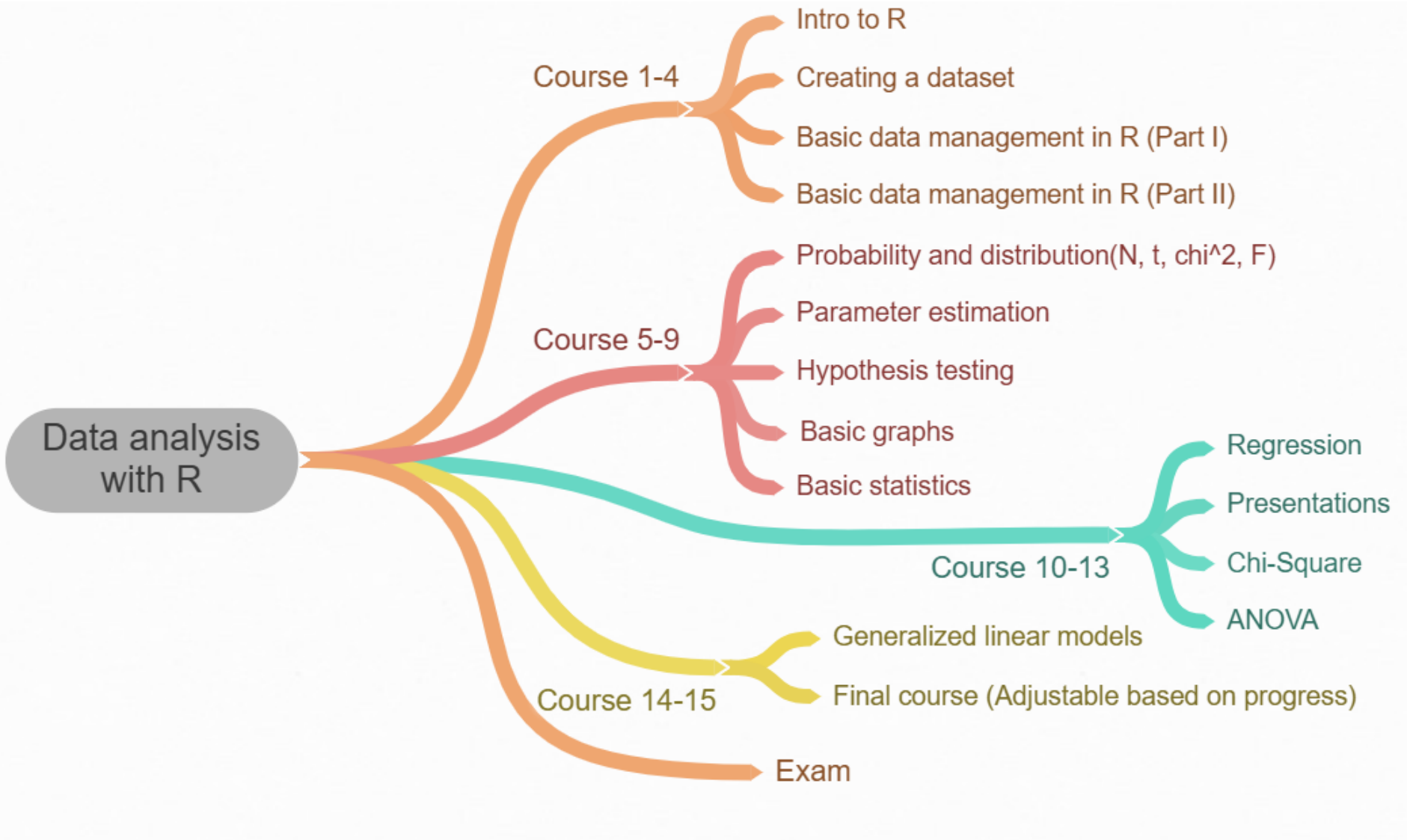
Data analysis with R

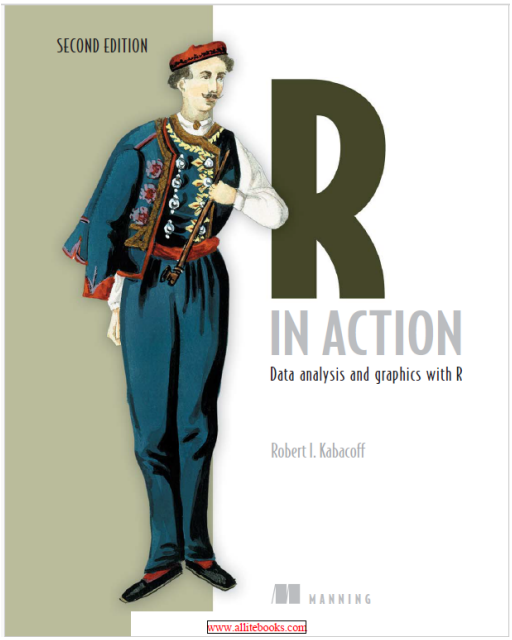
2.21.2025

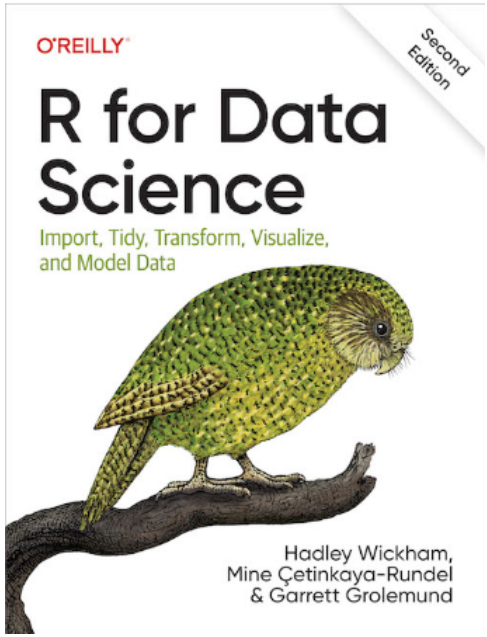
Roadmap

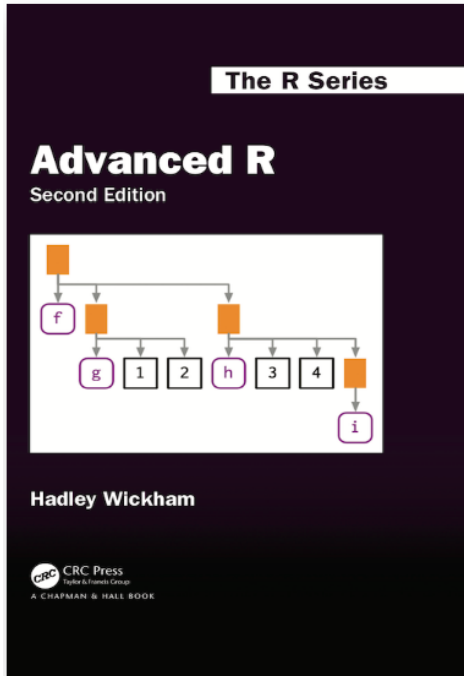

Data analysis with R
Course 1-4
Intro to R
Creating a dataset
Basic data management in R (Part I)
Basic data management in R (Part II)
Course 5-9
Probability and distribution(N, t, chi^2, F)
Parameter estimation
Hypothesis testing
Basic graphs
Basic statistics
Course 10-13
Regression
Presentations
Chi-Square
ANOVA
Course 14-15
Generalized linear models
Final course (Adjustable based on progress)
Exam

Recommended reading materials:

- "R in Action" by Robert Kabacoff
- "R for Data Science" by Hadley Wickham and Garrett Grolemund
- "Advanced R" by Hadley Wickham







Happy reading and learning!

Exam

- 40% coursework
- 60% open-note exam

This course

- A facilitator
- Voice your opinions, share your perspectives, engage in open discussions without fear of judgment

1 Preface: about data analysis

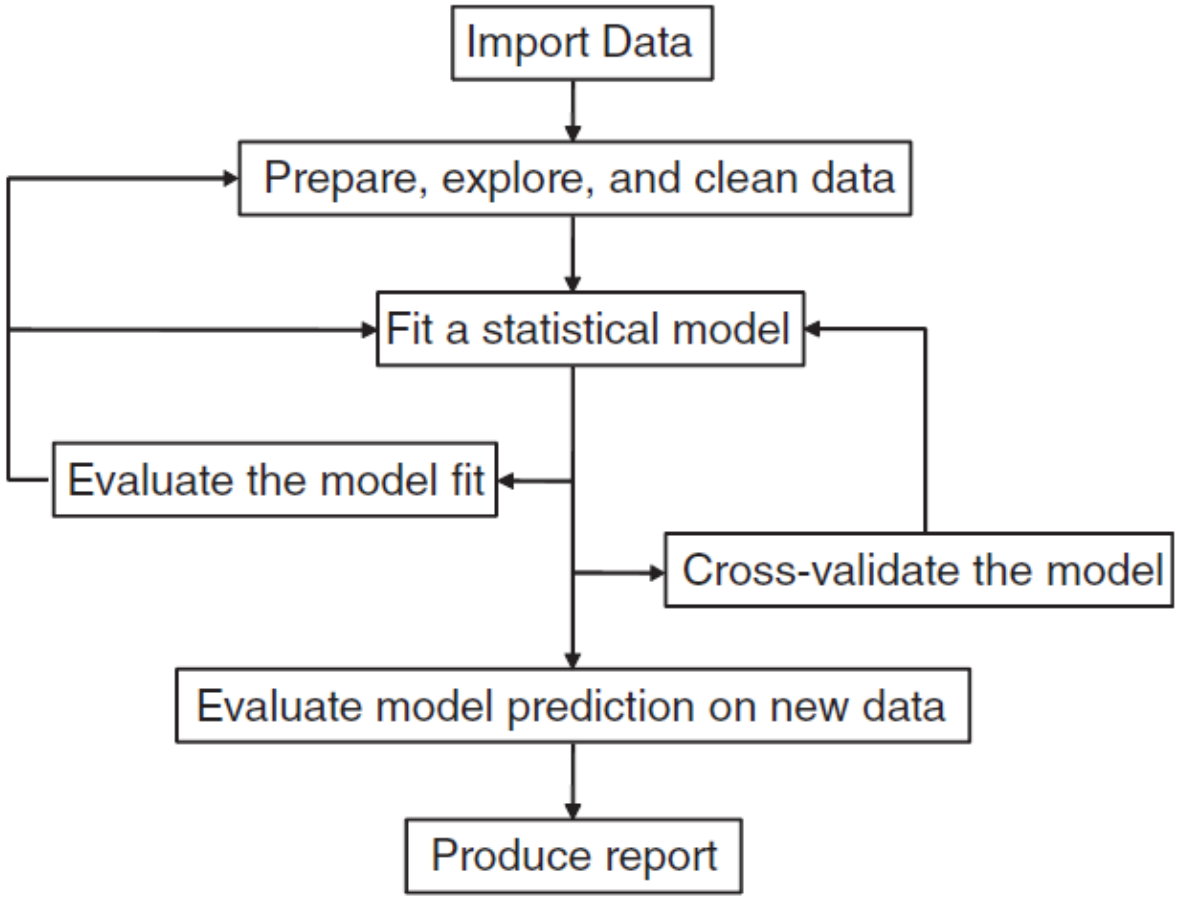
2 Obtaining and installing R

3 Working with R

4 Packages

5 Using output as input: reusing results

An example of a typical data analysis

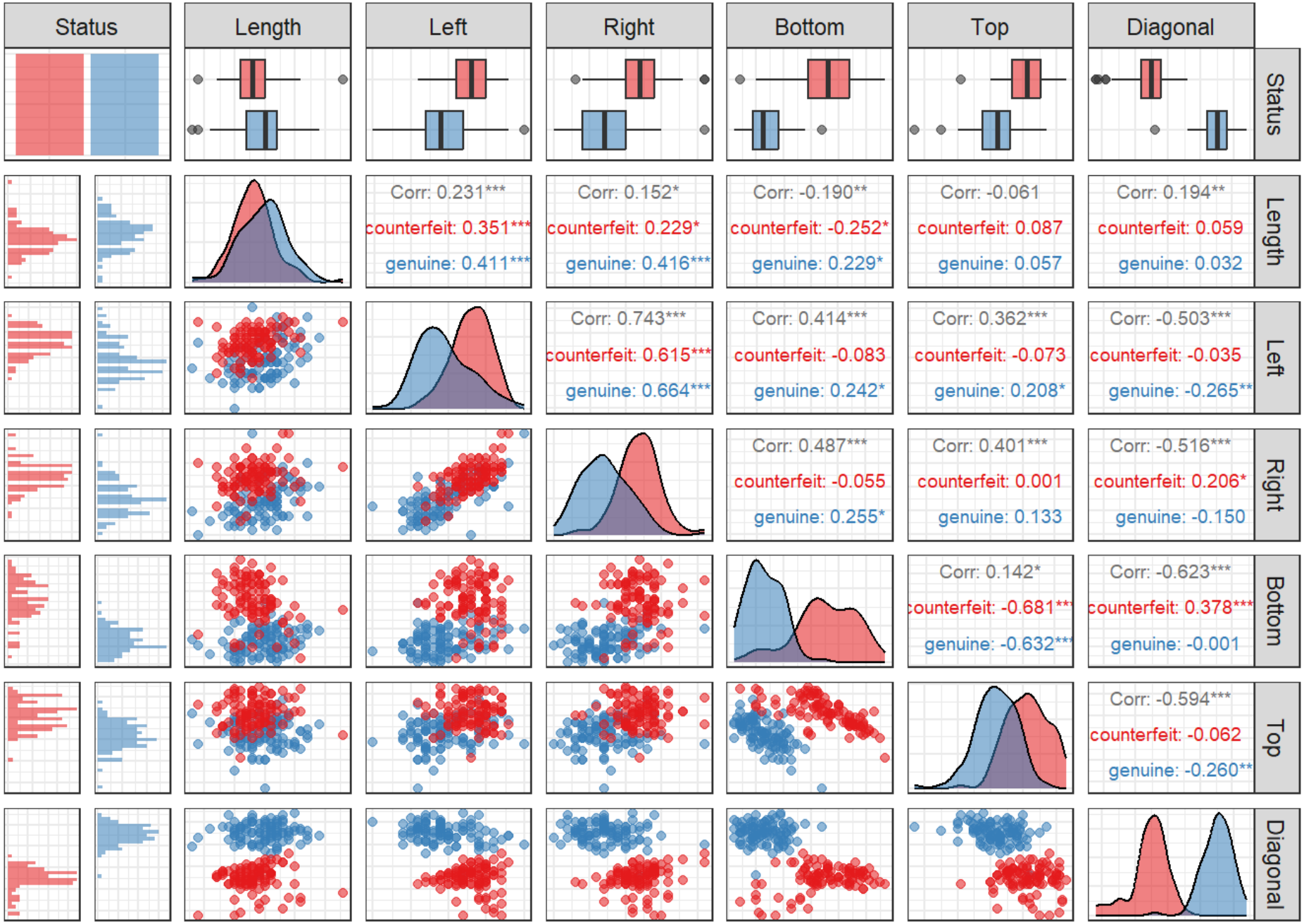


Data analysis nowadays

- Visual presentations: A picture really can be worth a thousand words
- A wide range of data sources (database management systems, text files, statistical packages, etc.): merge, clean, annotate, analyze, and present.

Why use R?

- R is free
- R is a comprehensive statistical platform, offering all manner of data-analytic techniques
- R contains advanced statistical routines not yet available in other packages
- State-of-the-art graphics capabilities
- A powerful platform for interactive data analysis and exploration
- R can easily import data from a wide variety of sources
- R functionality can be integrated into applications written in other languages, including C++, Java, Python, PHP, Pentaho, SAS, and SPSS
- R is likely to run on any computer you may have


Status
Length
Left
Right
Bottom
Top
Diagonal
Corr: 0.231***
counterfeit: 0.351***
genuine: 0.411***
Corr: 0.152*
counterfeit: 0.229*
genuine: 0.416***
Corr: -0.190**
counterfeit: -0.252*
genuine: 0.229*
Corr: -0.061
counterfeit: 0.087
genuine: 0.057
Corr: 0.194**
counterfeit: 0.059
genuine: 0.032
Corr: 0.743***
counterfeit: 0.615***
genuine: 0.664***
Corr: 0.414***
counterfeit: -0.083
genuine: 0.242*
Corr: 0.362***
counterfeit: -0.073
genuine: 0.208*
Corr: -0.503***
counterfeit: -0.035
genuine: -0.265**
Corr: 0.487***
counterfeit: -0.055
genuine: 0.255*
Corr: 0.401***
counterfeit: 0.001
genuine: 0.133
Corr: -0.516***
counterfeit: 0.206*
genuine: -0.150
Corr: 0.142*
Corr: -0.623***
counterfeit: 0.378***
genuine: -0.001
Corr: -0.594***
counterfeit: -0.062
genuine: -0.260**

Challenges

- R can have a steep learning curve

This course

- We'll tour the many features of R, covering enough material to get you started on your data, with pointers on where to go when you need to learn more.

1 Preface: about data analysis

2 Obtaining and installing R

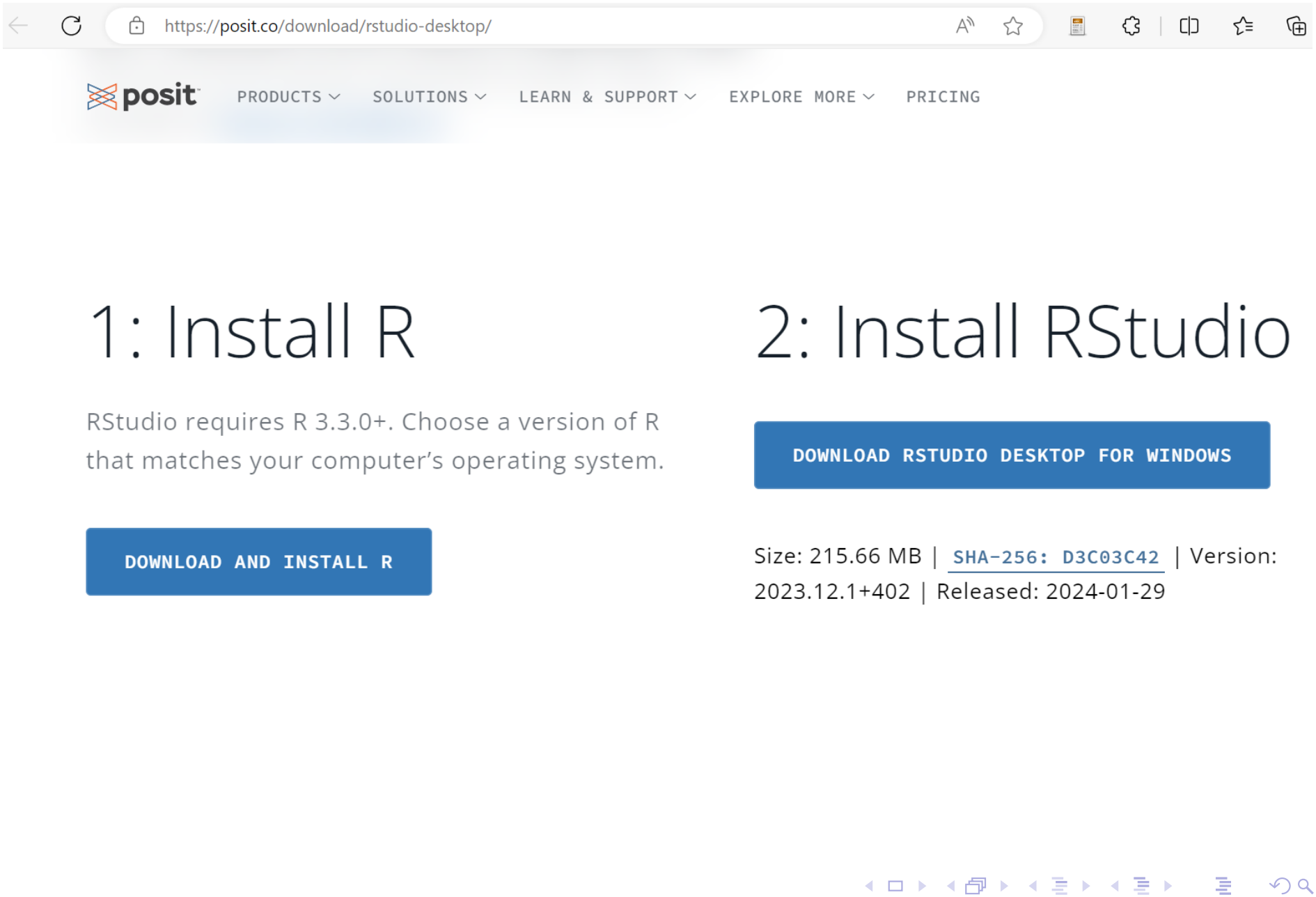
3 Working with R

4 Packages

5 Using output as input: reusing results

Obtaining and installing R

- R is freely available from the Comprehensive R Archive Network (CRAN) at http://cran.r-project.org.
- https://posit.co/download/rstudio-desktop/


https://posit.co/download/rstudio-desktop/
posit
PRODUCTS
SOLUTIONS
LEARN & SUPPORT
EXPLORE MORE
PRICING
1: Install R
RStudio requires R 3.3.0+. Choose a version of R that matches your computer's operating system.
DOWNLOAD AND INSTALL R
2: Install RStudio
DOWNLOAD RSTUDIO DESKTOP FOR WINDOWS
Size: 215.66 MB | SHA-256: D3C03C42 | Version: 2023.12.1+402 | Released: 2024-01-29

1 Preface: about data analysis

2 Obtaining and installing R

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Getting started

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

1721.R | Untitled1*

Source on Save | Run | Source

```
1 age <- c(1, 3, 5, 2, 11, 9, 3, 9, 12, 3)
2 weight <- c(4.4, 5.3, 7.2, 5.2, 8.5, 7.3, 6.0, 10.4, 10.2, 6.1)
3 mean(weight)
4 sd(weight)
5 cor(age, weight)
6 plot(age, weight)
```

6:18 (Top Level) R Script

Console | Terminal | Background Jobs

R 4.2.2 · ~/

```
> age <- c(1, 3, 5, 2, 11, 9, 3, 9, 12, 3)
> weight <- c(4.4, 5.3, 7.2, 5.2, 8.5, 7.3, 6.0, 10.4, 10.2, 6.1)
> mean(weight)
[1] 7.06
> sd(weight)
[1] 2.077498
> cor(age, weight)
[1] 0.9075655
> plot(age, weight)
>
```

Environment | History | Connections | Tutorial

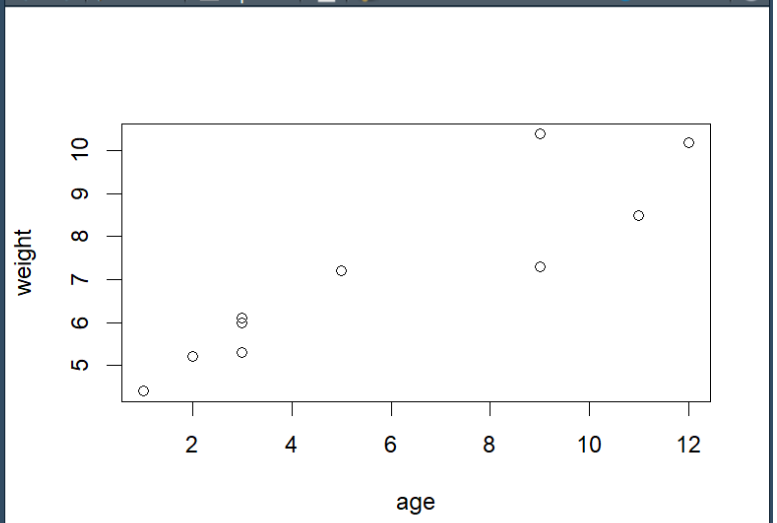
Import Dataset | 1.39 GiB | List

R | Global Environment

aovmodel_y1...	List of 13
c1721	33517 obs. of 5 variables
c2017	12582 obs. of 5 variables
c2018	12787 obs. of 5 variables
c2021	8148 obs. of 5 variables

Files | Plots | Packages | Help | Viewer | Presentation

Zoom | Export | Publish



The interface

- Source Code Editor (Top Left): This is where you write your R scripts and functions.
- Console (Bottom Left): R's interactive window where you can directly enter R commands and see the output immediately.
- Environment/History (Top Right): the objects in the current workspace, such as data frames, vectors, functions, etc.
- Files/Plots/Packages/Help (Bottom Right)

Manage the R workspace

Function	Action
getwd()	Lists the current working directory.
setwd("mydirectory")	Changes the current working directory to mydirectory.
save.image("myfile")	Saves the workspace to myfile (default = .RData).
load("myfile")	Loads a workspace into the current session.

An example of commands used to manage the R workspace

```
getwd()
setwd("C:/myprojects/project1")
```

1 Preface: about data analysis

2 Obtaining and installing R

3 Working with R

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5 Using output as input: reusing results

Packages

- Packages are collections of R functions, data, and compiled code in a well-defined format.
- The directory where packages are stored on your computer is called the library.
- R comes with a standard set of packages (including base, datasets, utils, grDevices, graphics, stats, and methods).
- Other packages are available for download and installation.

http://cran.r-project.org/web/packages.

Packages

Installing a package

1. To install a package for the first time, use the install.packages() command.
2. You only need to install a package once.
3. Use the command update.packages() to update any packages that you've installed.

Packages

Loading a package

- To use a package in an R session, you need to load the package using the library() command.
- You must have installed a package before you can load it.
- You'll only have to load the package once in a given session.

Packages

Learning about a package

- Getting help: Using help() with any of function or dataset names provides further details.

Common mistakes in R programming

If your program generates an error, be sure to check for the following:

- R is case sensitive: help() will work but Help() and HELP() not.
- install.packages("gclus") works, whereas install.packages(gclus) generates an error.
- Using the "\" in a path name on Windows. setwd("c:\mydata") generates an error. Use setwd("c:/mydata") or setwd("c:\\mydata") instead.
- Using a function from a package that's not loaded.

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A sample R session

RStudio

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1721.R | Untitled1*

Source on Save | Run | Source

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Console | Terminal | Background Jobs

R 4.2.2 · ~/

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Environment | History | Connections | Tutorial

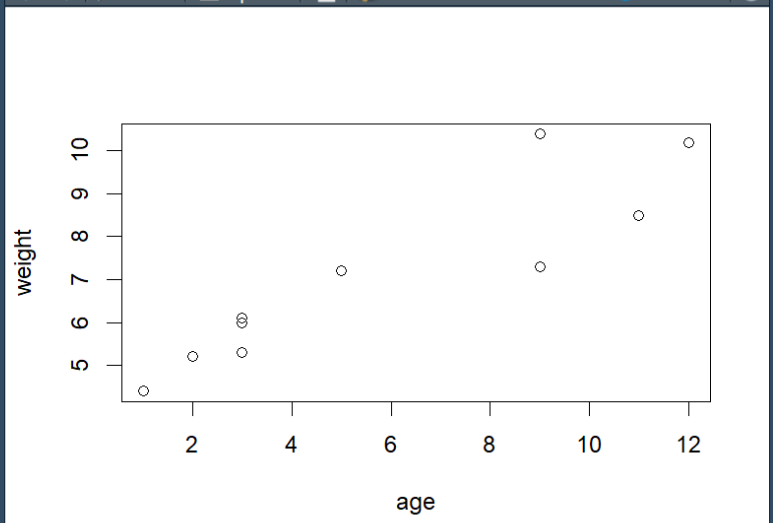
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Files | Plots | Packages | Help | Viewer | Presentation

Zoom | Export | Publish



Using output as input: reusing results

Assignment operator

variable name <– value

Note: The <– operator is preferred in R over the = operator for assignment

- Assigning a single value:
  x <– 10
- Assigning a vector:
  myvector <– c(1, 2, 3, 4, 5)
- Assigning a matrix:
  mymatrix <– matrix(1:9, nrow = 3)

Using output as input: reusing results

- The output of analyses can easily be saved and used as input to additional analyses.

An example

1. We want to predict how car weight (wt) influences miles per gallon (mpg)
2. One of the datasets that comes preinstalled with R: mtcars
3. Run a simple linear regression
4. Store the results in an object, including the predicted values, residuals, regression coefficients and more

Although no output is sent to the screen, the results can be both displayed and manipulated further.

Working through an example

The task

1. Check the current working directory
2. Change the location
3. Open the general help, and look at the "Introduction to R" manual
4. Install the "ggplot2" package (a package for creating elegant data visualisations) and "ggplot2movies"
5. Load them
6. Read the description of the dataset "movies"
7. Run the example that comes with the dataset
8. Save the workspace to file "course1.RData"

Working through an example

Solutions

1. getwd()
2. setwd("") or use the menu
3. Use the help menu (bottom right)
4. install.packages("ggplot2"), install.packages("ggplot2movies")
5. library(ggplot2),library(ggplot2movies)
6. help(movies)
7. dim(movies), head(movies)
8. save.image(file = "Course1.RData")